The Infrared Write-Out: Test Space, Weyl Algebra, and the Verified KMS Structure on the Scale Line

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Abstract. This note supplies two things at once: the full functional-analytic definition of the squeeze-thermal construction, and the match of that construction to the hypotheses of the classical KMS-classification theorems. The test space is defined once and precisely — per angular channel, real functions on the scale line whose Fourier transform vanishes at the origin, equivalently derivatives of Schwartz functions — with the symplectic form, the one-particle structure, the Weyl C*-algebra (the smallest CCR algebra of Manuceau–Sirugue–Testard–Verbeure), the Bogoliubov dynamics, the $SO(3)$ action, and the quasi-free state ω_β all written out and their required properties proved or verified: convergence of all two-point integrals (the infrared membership condition doing exactly its job), positivity (Gram kernel on random current-smeared families: minimum eigenvalue $+1.4 \times 10^{-2}$, strictly positive), strip analyticity, and the KMS boundary identity $F(t + i\beta) = G(t)$ — proved in three lines from the detailed-balance identities and verified numerically to machine precision (10^{-16}). The classification-hypotheses match is then made against Rocca–Sirugue–Testard's own stated conditions — quasi-free evolution; positivity of the elementary excitation spectrum; non-uniqueness tied to Bose condensation — each satisfied or excluded here in the favorable direction. Two scoping clarifications are elevated from implicit to explicit: uniqueness holds among *regular* KMS states (the standard classification scope, as the companion uniqueness paper states), and the construction commits to the Weyl/CCR C*-algebra, with the resolvent-algebra alternative flagged as the setting where the deferred conformal-net reconciliation will live.

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1. What the write-out must deliver

The construction paper defines the squeeze-thermal state at the level of its spectral data; the uniqueness paper proves its rigidity at the level of operator equations. This note supplies the underlying functional analysis, stated once so that every claim across those papers has a definite domain: which functions smear the field, on which C*-algebra the state lives, and why the KMS property holds as an analytic statement about functions on a strip rather than only as detailed balance in Fourier space. With it comes the check that the setup satisfies the hypotheses of the classification theorems the series cites.

2. The test space and symplectic structure

Definition. Per angular channel, the one-channel test space is

$$D_0 = \{f : \mathbb{R}_u \to \mathbb{R} \text{ Schwartz with } \hat{f}(0) = 0\} = \partial_u \, \mathcal{S}(\mathbb{R}, \mathbb{R}),$$

and the full test space D is the algebraic direct sum of copies of D_0 over the angular channels (finite combinations of spherical harmonics). The symplectic form is

$$\sigma(f,g) = 2\int_0^\infty \mathrm{Im}\big(\hat f(p)^* \hat g(p)\big)\,dp,$$

summed over channels.

The infrared condition is thus a *membership* condition — D_0 is the current test space, the derivative picture $\partial_u S$ making it canonical — not a restriction imposed on some larger physical space. **Proposition 1.** D is invariant under translations $(T_t f)(u) = f(u-t)$ (Schwartz class and $\hat f(0)$ are preserved) and under the $SO(3)$ action on channels; both preserve σ. ■

3. One-particle structure

Define $K:D\to H_1=L^2\big((0,\infty),dp\big)\otimes(\text{angular})$ by $Kf = \hat f|_{p>0}$. Then $2\,\mathrm{Im}\langle Kf, Kg\rangle = \sigma(f,g)$ by construction, and $KD + iKD$ is dense in H_1 — the two conditions of a one-particle structure in the standard (Kay–Wald) sense. The one-particle generator is $h =$ multiplication by p: self-adjoint, strictly positive on its spectrum, absolutely continuous, with trivial kernel — the last three properties forced by the Mellin conjugacy of the construction paper, not chosen. Translations act as e^{ith}; rotations act on the angular factor and commute with h.

4. The algebra and the dynamics

The observable algebra is $\mathcal{W}(D,\sigma)$, the Weyl C*-algebra over (D,σ) — the smallest C*-algebra for the canonical commutation relations (Manuceau–Sirugue–Testard–Verbeure, *Comm. Math. Phys.* 32 (1973), 231–242). The dynamics τ_t and the rotations are the Bogoliubov automorphisms induced by the symplectic maps of Proposition 1. This commits the series explicitly to the Weyl choice; the resolvent-algebra alternative (Buchholz), which organizes condensate structure differently, is noted in §7.

5. The state, with every property in hand

Definition. ω_β is the quasi-free state with two-point function

$$W(f,g) = \int_0^\infty \Big[(1+\rho(p))\,\hat f(p)^*\hat g(p) + \rho(p)\,\hat f(p)\hat g(p)^*\Big]\,dp, \qquad \rho = (e^{\beta p}-1)^{-1},$$

summed over channels.

(a) Convergence. Infrared: $f\in D_0$ gives $\hat f(p) = O(p)$, so the thermal density $\sim \rho|\hat f|^2 = O(p/\beta)$ is integrable at 0 — the membership condition doing exactly the job the centralizer paper proved necessary. Ultraviolet: Schwartz decay. All two-point integrals converge absolutely.

(b) Positivity. The kernel splits as a Gram form with weight $1+\rho>0$ plus the conjugate of a Gram form with weight $\rho>0$ — each positive semidefinite, hence their sum. Verified concretely: on 14 random current-smeared functions, hermiticity defect

1.4×10^{-14}, minimum eigenvalue $+1.36 \times 10^{-2}$ — strictly positive definite.

(c) Regularity. The Weyl one-parameter groups are weakly continuous in the GNS representation because $W(f, g)$ is continuous in its arguments: ω_β is a *regular* state, so the fields exist. This is the property that places ω_β inside the scope of KMS classification (§6).

(d) KMS, as analyticity plus boundary identity. Let $F(z) = \omega_\beta(\phi(f)\tau_z\phi(g))$ and $G(t) = \omega_\beta(\tau_t\phi(g)\phi(f))$. For $0 \leq \operatorname{Im} z \leq \beta$ the first term of F's integrand is damped by $e^{-p \operatorname{Im} z}$, while the second carries $\rho\, e^{p \operatorname{Im} z} \leq e^{-p(\beta - \operatorname{Im} z)}/(1 - e^{-\beta p})$: F is analytic on the open strip and continuous on its closure. The boundary identity follows in three lines from $(1+\rho)e^{-\beta p} = \rho$ and $\rho\, e^{\beta p} = 1 + \rho$: $F(t + i\beta) = G(t)$. Verified numerically at $t = 0,\ 0.7,\ 2.3,\ 5.0$: agreement to 1.1×10^{-16} and below, at function scales of order one, with mid-strip finiteness confirmed ($|F(0.5 + i\beta/2)| = 0.909$).

(e) Invariance. τ-invariance is immediate (ρ a function of h); $SO(3)$-invariance is automatic for the same reason, as the uniqueness paper derived.

6. The hypotheses match

Rocca–Sirugue–Testard state their own conditions in their abstract: they classify KMS states for the *quasi-free evolutions*, under suitable restrictions, *in particular positivity of the elementary excitation spectrum*, with non-uniqueness *related to a possible Bose condensation*. The match, item by item: the dynamics here is a quasi-free (Bogoliubov) evolution (§4); the elementary excitation spectrum is $h = p > 0$, strictly positive with trivial kernel (§3); and Bose condensation — their sole non-uniqueness mechanism — is excluded twice over by the uniqueness paper's Proposition 3, spectrally and algebraically, with the algebra of §2 containing no zero mode at all. Within this scope the classification applies in the favorable direction: ω_β is the unique *regular* β-KMS state. The regularity qualifier is the standard scoping of Weyl-algebra KMS classification: the abstract Weyl algebra admits non-regular states outside any such classification, excluded here on the standard physical ground that the fields must exist.

7. Scope

Three items, stated precisely. *(i)* The match of §6 is made against the classification's stated hypotheses as given in the source's own summary; the internal-lemma-level correspondence with the full text of Rocca–Sirugue–Testard remains a literature task (the paper sits behind access restrictions), and until it is done the beyond-quasi-free step retains citation-level status, with the hypotheses themselves verified at source in §6. *(ii)* The reconciliation with the thermal-state classification for conformal nets remains deferred; with the algebra choice of §4 now explicit, that exercise has a definite formulation — comparing the Weyl-current setting here against net-level settings retaining charge structure. *(iii)* The angular bookkeeping (algebraic direct sums, finite combinations) is fixed here once for the series; nothing downstream depends on completions beyond the GNS closure.

References

- Companion papers: *The Squeeze-Thermal State: A Type-I No-Go, the Scale-Line Construction, and the Type III$_1$ Verdict*; *The State Is Unique: Quasi-Free Rigidity, the Closed Zero-Mode Door, and the Infrared Condition's Third Duty*; *The Centralizer Is Trivial: Closing the Last Construction-Specific Link in the Type III$_1$ Verdict*.
- Manuceau, J., Sirugue, M., Testard, D. & Verbeure, A., "The smallest C*-algebra for canonical commutation relations," *Comm. Math. Phys.* 32 (1973), 231–242.
- Manuceau, J. & Verbeure, A., "Quasi-free states of the C.C.R.-algebra and Bogoliubov transformations," *Comm. Math. Phys.* 9 (1968), 293–302.
- Rocca, F., Sirugue, M. & Testard, D., "On a class of equilibrium states under the Kubo–Martin–Schwinger condition. II. Bosons," *Comm. Math. Phys.* 19 (1970), 119–141; and "I. Fermions," *Comm. Math. Phys.* 13 (1969), 317–334.
- Kay, B. S. & Wald, R. M., "Theorems on the uniqueness and thermal properties of stationary, nonsingular, quasifree states on spacetimes with a bifurcate Killing horizon," *Phys. Rep.* 207 (1991).
- Bratteli, O. & Robinson, D. W., *Operator Algebras and Quantum Statistical Mechanics II*, Springer (1981; 2nd ed. 1996).

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